



2026

THE 2026 GLOBAL OPEN SOURCE DEVELOPMENT REPORT

PREVIEW EDITION

CC  SI





CAST UN Consultative Committee on Open Source Innovation

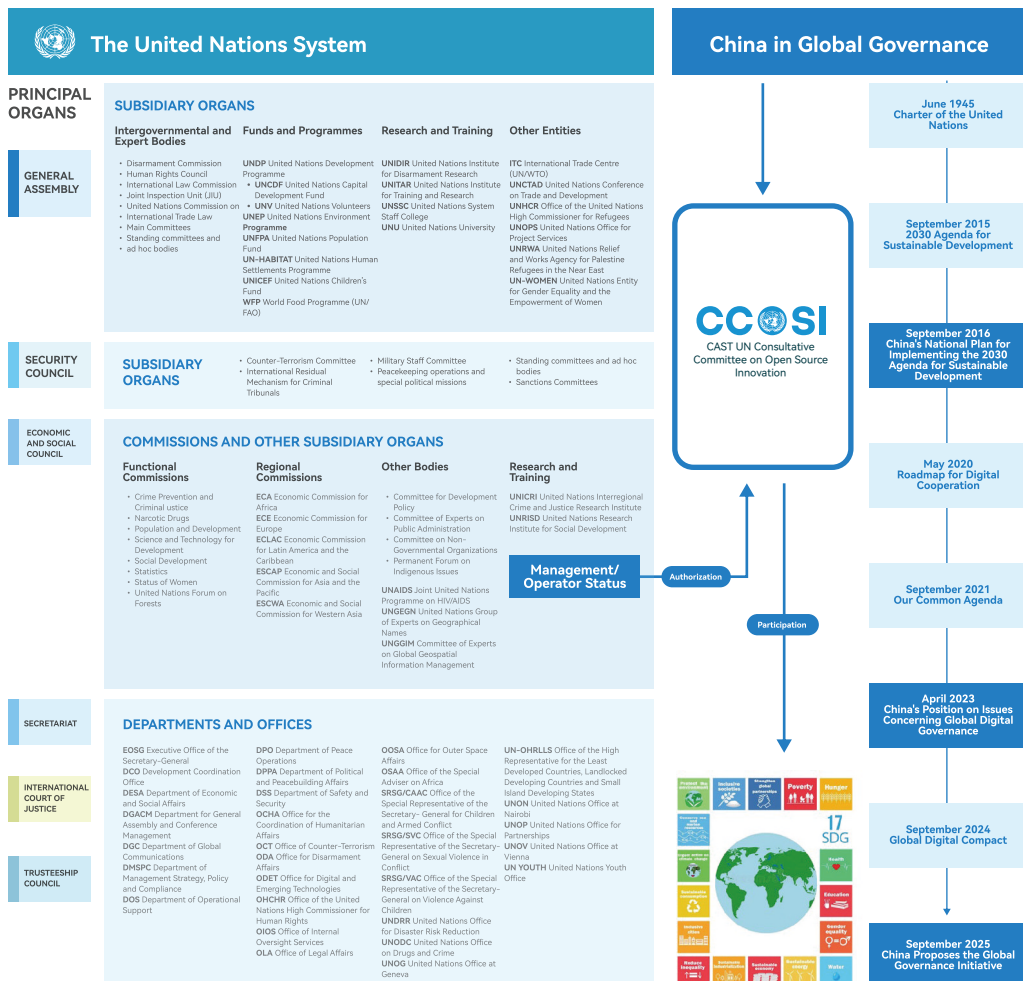


CAST UN Consultative Committee on Open Source Innovation (CCOSI)

**Group Photograph for the Seminar Concerning Sustainable
Development of Open-Source Artificial Intelligence**

Background and Approving Organization

The **China Association for Science and Technology (CAST) United Nations Consultative Committee on Open Source Innovation (CCOSI)** was established with the approval of the China Association for Science and Technology. Relying on the special consultative status of CAST with the United Nations Economic and Social Council (ECOSOC), CCOSI serves as an important mechanism for promoting international cooperation in the field of open source innovation under the global framework of science and technology governance. The committee's secretariat is managed by the China Communications Standards Association (CCSA). The primary aim of CCOSI is to organize domestic and international experts to deeply engage in agenda-setting and policy dialogue on open source innovation within the United Nations and related international organizations.



Core Functions and Responsibilities

The committee is primarily responsible for organizing research conducted by experts in related fields, participating in meetings and events organized by the United Nations and relevant international organizations, carrying out international cooperation, hosting regular thematic forums, and producing policy recommendations or research reports. The secretariat is responsible for providing necessary office space, personnel, and equipment support, while also coordinating the daily operations of the committee. Additionally, the committee actively mobilizes the institutions of its members to leverage their resource advantages to support consultative activities and works in collaboration with the CAST United Nations Consultative Office to provide intellectual support and organizational assurance for China's participation in global science and technology governance.

Basic Rights

- Obtain a UN pass and access UN premises
- Can send 5 representatives each to institutions and permanent missions in the UN buildings in New York, Geneva, and Vienna
- Understand and participate in ECOSOC's (Economic and Social Council) temporary agenda
- Attend public international conferences and activities
- Make written or oral statements at international conferences and activities
- Organize and participate in parallel activities related to the main activities organized by ECOSOC's functional commissions
- Establish network connections and lobbying groups

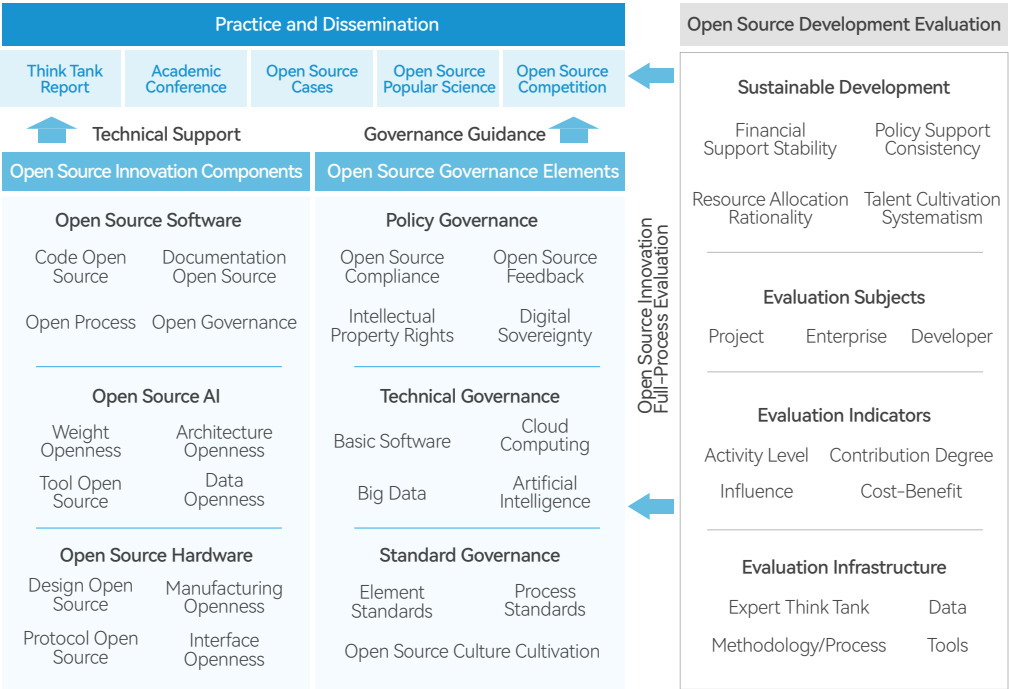
Other Rights

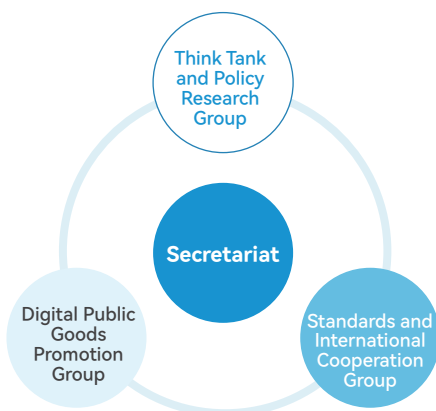
- Conduct scientific analysis of relevant issues based on its professional experience
- Serve as a UN early warning mechanism
- Assist in monitoring and implementing international agreements
- Help raise public awareness of relevant issues
- Assist in promoting the achievement of the UN's goals and purposes
- Provide important information for UN organizational activities

Organizational Structure and Operational Mechanism

The committee has established a Chairman’s Office as the decision-making body and has set up a secretariat and several professional working groups, including the Think Tank and Policy Research Group, Digital Public Goods Promotion Group, and Standards and International Cooperation Group. These working groups focus on systematic work in areas such as global open source trend analysis, internationalization of open source technology standards, and the incorporation of open source outcomes into the global digital public goods framework.

The secretariat, as the operational hub, is responsible for agenda management, member liaison, meeting organization, budget coordination, information aggregation, and cross-group project advancement, ensuring the efficient, collaborative, and orderly execution of the committee’s strategic objectives.





Strategic Positioning and Development Goals

The Committee is positioned as a professional platform representing China in global open-source governance, dedicated to building an open and inclusive international science and technology cooperation system. Its vision is to promote global scientific openness and sharing, contributing to the sustainable development of human society. Through the United Nations consultative channels, the Committee serves national strategic needs, disseminates China's practices and ideas in open-source innovation, and brings together global forces to drive the co-construction and sharing of a global innovation ecosystem.

International Cooperation and Global Participation

The Committee actively engages with foreign experts, facilitates international guest exchanges to China, and carries out international collaboration projects such as the "Global Open Source Development Status Survey" and "Global Digital Public Goods Catalog Construction." Additionally, by participating in international open-source community activities and publishing declarations such as the "**Open Source Innovation International Cooperation Declaration**," the Committee aims to enhance China's voice and influence within the global open-source governance system, fostering the development of an open, inclusive, and collaborative global science and technology community.

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EXECUTIVE SUMMARY

As digital transformation reshapes the global economic, social, and environmental landscape, open source has evolved from software development practice into critical infrastructure for global innovation collaboration and sustainable development. Its characteristics—open sharing, collective governance, and non-exclusivity—endow it with strong public good attributes, enabling systemic impact in knowledge dissemination, technology inclusion, and capacity building. The 2026 Global Open Source Development Report aims to systematically map the current state and trends of the global open source ecosystem, assess its tangible contributions to the United Nations Sustainable Development Goals (SDGs), and provide data-driven insights for policymakers, enterprises, developer communities, and international organizations.

Centered on the principle of “Open Source for Good”, the report introduces the Open Source Development Goals (OSDGs) framework, translating the abstract SDGs into actionable pathways within the technology ecosystem. Built on four pillars—Vitality, Contribution, Connection, and Evolution—the OSDGs framework establishes a quantifiable assessment system across dimensions such as developer engagement, collaboration depth, technical impact, green innovation, and equitable technology. It provides a unified benchmark for evaluating the health and societal value of open source projects, while clarifying their strategic role in advancing inclusive innovation and bridging the digital divide. The report follows a logical structure of “Goals—Observation—Assessment—Insight—Special Study”:

- **Chapter 1: Open Source Development Goals (OSDGs)** defines the role of open source as a digital public good and maps its alignment with the 17 SDGs;
- **Chapter 2: Global Open Source Observatory (GOSO)** presents five dynamic maps—activity, contribution, technology, talent, and academia, to visualize the global distribution and collaboration networks of open source, revealing regional disparities and emerging potential;
- **Chapter 3: Global Open Source Influence Ranking (GOSR)** establishes a comparative influence framework across five dimensions—jurisdictions, enterprises, technology domains, projects, and emerging initiatives—reflecting the evolving roles of different stakeholders;
- **Chapter 4: Global Open Source Technology Landscape (GOSL)** examines key technical domains, including AI and large models, big data, cloud infrastructure, database systems, and the PyTorch Foundation ecosystem, analyzing technological evolution, competitive dynamics, and collaboration models;
- **Chapter 5: State of Open Source AI (OSAI)** responds to the rapid advancement of AI by proposing an openness grading standard for large models and a global identification initiative, while analyzing innovation trends and governance challenges through patent and publication data.

This report is based on the Information Technology – Open Source Governance series of standards issued by the Ministry of Industry and Information Technology of the People's Republic of China. It integrates international best practices and references the LF CHAOSS community health metrics to establish an

indicator system covering activity, contribution volume, contribution diversity, and impact. Additionally, the appendix introduces the Global Digital Public Goods Classification and Identification Framework, providing a unified identity and classification system for open-source software, open data, AI models, and open standards, thereby enhancing resource discoverability, interoperability, and policy alignment.

The report finds that open source is becoming a key enabler of the SDGs: lowering access barriers to quality education, supporting climate modeling and energy efficiency in environmental protection, enhancing transparency and public trust in governance, and promoting technological democratization to empower developing countries in global digital innovation.

Yet, the potential of open source is not self-realizing. Challenges such as growing geopolitical fragmentation, sustainability gaps in core project maintenance, and insufficient security and ethical governance remain. The report calls for stronger multi-stakeholder collaboration and improved institutional design to shift open source from spontaneous collaboration toward systematic governance. Only by upholding the principles of openness, transparency, and inclusivity—and mobilizing joint efforts from governments, industry, academia, and civil society—can open source continue to serve the shared well-being of humanity.

CONTENTS

The background of the page is a vibrant blue watercolor wash. It features soft, blended strokes of various shades of blue, from a deep cerulean at the top to a lighter, almost white turquoise at the bottom. There are numerous small, darker blue speckles and larger, more saturated patches scattered throughout, giving it a textured, organic feel. The overall effect is a calm yet dynamic aquatic or sky-like pattern.

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PREFACE

This report takes the Declaration on International Open Source Innovation Cooperation as its ideological foundation. The Declaration embodies a shared commitment from governments, enterprises, academia, and communities worldwide towards a future of open-source co-governance. As the first comprehensive key document fully responding to this Declaration, the Global Open Source Development Report 2026 aims to track how the global open-source ecosystem implements core principles such as "connection, security, inclusivity, green development, and co-governance" through data and case studies. It also provides technological pathways and practical references for achieving the United Nations' 2030 Sustainable Development Goals.

Declaration on International Open Source Innovation Cooperation

Open Source Co-Creation for the Future: Towards Global Digital Co-Governance

I. A Call of Our Time: Advancing into a New Era of Digital Civilization

We stand at a pivotal moment in humanity's transition towards a digital era. Technological revolutions are reshaping the world with unprecedented depth and breadth, and open source, as the most vibrant collaboration paradigm of the digital age, has become a key engine driving scientific and technological innovation, social progress, and sustainable development.

However, the global digital divide continues to widen, risks of technological fragmentation are intensifying, and ethical challenges surrounding artificial intelligence are becoming increasingly prominent. Faced with common challenges such as climate change, public health, and educational equity, only by adhering to openness, collaboration, and co-governance can we ensure that technology serves good and benefits all of humanity.

Hereby, we—the representatives from governments, technology enterprises, research institutions, open-source communities, developers, and civil society around the world—solemnly issue this Declaration. We call for guidance by the spirit of openness and a path of global collaboration to jointly build and share a fairer, safer, and more sustainable digital future.

This Declaration proposes a global open-source action framework: rooted in the cornerstone of digital civilization, supported by pillars of technological governance, and driven by the mission of social value realization. It constructs a complete logical chain from connection to co-governance, positioning open source as an accelerator for achieving the United Nations' 2030 Sustainable Development Goals.

II. Shared Vision: Building a Co-Governed and Shared Digital Civilization

We are committed to building an open, inclusive, secure, and sustainable global open-source innovation ecosystem, ensuring that technology truly serves the well-being of all humanity.

We firmly believe: Open source is not merely the sharing of code, but also the establishment of trust, the convergence of wisdom, and the shared responsibility. Only by adhering to collaborative principles of diverse participation, transparent governance, and contribution-oriented practices can technological innovation become a collective force for peace, prosperity, and sustainable development.

We collectively advocate incorporating open-source innovation into global digital public goods strategies, positioning it as a key pathway to achieve the *2030 Agenda for Sustainable Development*, thereby helping to construct a co-governed, shared, and mutually beneficial digital civilization.

III. Core Principles: Action Framework

First Layer: Fortifying the Cornerstone of Digital Civilization

- **Principle One: Uphold Open Connection**

"Interconnectedness, Inclusive Coexistence"

Promote the inclusive development of digital infrastructure and support capacity building in developing countries. Eliminate barriers of language, geography, and capability; encourage multilingual development environments and localized collaboration tools. Adhere to non-discrimination and inclusiveness in technological development, ensuring everyone can freely and equally participate in open source, making technological connection a bridge for civilizational advancement.

- **Principle Two: Uphold Security and Trustworthiness**

"Digital Ecosystem, Foundation of Trust"

Oppose internet fragmentation and uphold global network interconnectivity. Safeguard the neutrality and stability of critical infrastructure and core resources; strengthen the security of open-source supply chains. Build a trustworthy, resilient, and sustainable technological ecosystem, solidifying the security baseline of global digital civilization.

Second Layer: Strengthening Pillars of Technological Governance

- **Principle Three: Uphold Data Ethics**

"Algorithms for Good, Privacy First"

Uphold data security, privacy protection, and algorithmic fairness, guarding against technological misuse and discrimination. Promote AI development guided by principles of inclusivity, security,

cooperation, and agility. Encourage open source to serve as an experimental platform for AI explainability, auditability, and ethical practice, ensuring intelligent technologies always serve human dignity and rights.

- **Principle Four: Uphold Diversity and Inclusivity**

"Collective Intelligence, Shared Rights"

Respect diversity in culture, gender, age, and ability, opposing any form of technological exclusion and digital discrimination. Support the participation of marginalized groups in open source, promote barrier-free design and equitable access mechanisms, ensuring every voice is heard and every contribution respected.

- **Principle Five: Uphold Global Collaboration**

"Cross-Boundary Synergy, Global Interconnection"

View open source as a bridge connecting the world. Support cross-national, cross-institutional, and cross-domain collaborative development, breaking down geographical and institutional barriers. Promote knowledge flow, capability complementarity, and resource sharing, building a decentralized yet interconnected global innovation network, strengthening South-South cooperation and multilateral practices.

Third Layer: Co-Creating a Future of Social Value

- **Principle Six: Uphold Green Innovation**

"Green-Driven, Innovation for Longevity"

Prioritize support for open-source projects addressing climate change, environmental protection, and clean energy. Promote open source as a core implementation path for digital public goods, widely applied in carbon monitoring, energy optimization, and climate adaptation fields. Ensure technological development does not come at the expense of the environment, providing innovative support for global ecological security and sustainable development.

- **Principle Seven: Uphold Co-Governance and Sharing**

"Open Source for the Public, Shared Outcomes"

Promote the widespread application of open-source technology in government governance, public health, and educational equity, serving modernization of social governance and improvement of people's livelihoods. Encourage open source to be a carrier of digital public goods, establishing contribution-oriented incentive mechanisms so that innovative outcomes benefit everyone, especially marginalized groups.

IV. Action Consensus: Moving Towards Global Digital Co-Governance

We call upon governments, enterprises, academia, and civil society worldwide to join hands, using this Declaration as a basis for consensus, to implement the action framework of "connection, security, data, inclusivity, collaboration, green development, and co-governance," advancing open source as a model

practice for a "Global Digital Compact."

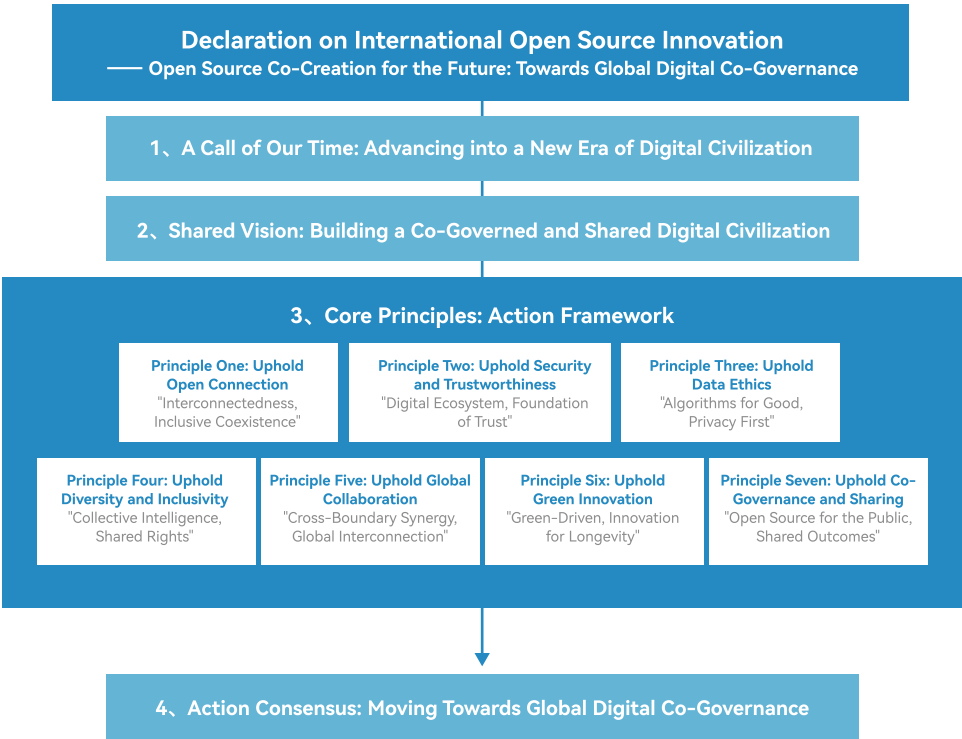
We support leveraging the United Nations, regional organizations, and multilateral mechanisms to deepen cooperation, consolidate consensus, and strive for early gains in urgent areas such as artificial intelligence, cyberspace, digital infrastructure, data governance, and climate technology.

We firmly believe: Open source innovation belongs to all humanity; the digital future is created by us together.

Conclusion

Open source is our steadfast adherence to the spirit of openness, our belief in human solidarity, and our shared pursuit of a better world. Let us replace division with collaboration, monopoly with sharing, and meet challenges with co-governance. Together, let us build a more open, fair, and sustainable digital civilization.

Open source innovation belongs to all humanity; the digital future is created by us together.



CHAPTER 1:

OPEN SOURCE DEVELOPMENT GOALS

Chapter 1: Open Source Development Goals

■ The Light of Open Source: Charting the Action Blueprint for a Sustainable Future

This chapter focuses on the strategic role of open-source technology in global sustainable development and introduces the **Open Source Development Goals (OSDGs)** analytical framework. The report highlights that open source, as a new form of global public good, possesses non-exclusivity, non-competitiveness, and strong positive externalities. It has become a key infrastructure supporting knowledge sharing, collaborative innovation, and inclusive capacity-building. By providing open software, data, and models, open source is lowering the barriers to technology access and driving equitable global development.

The OSDGs framework systematically maps the United Nations Sustainable Development Goals (SDGs) onto open-source practices, creating an evaluation system that spans three dimensions: **Environment (E)**, **Society (S)**, and **Governance (G)**. This framework aims to translate abstract sustainability goals into actionable technological pathways, measuring the contribution of open source in promoting green transformation, enhancing digital inclusion, and increasing system transparency, thus providing a basis for policy-making and resource allocation.

Critically, the OSDGs resonate with the global urgency for technology democratization, especially in cutting-edge fields such as artificial intelligence. Open source is becoming a core force to break monopolies and accelerate inclusivity. By establishing institutional solutions such as the "Open Source Large Model Grading Evaluation Standards," the OSDGs address the governance gaps and trust crises facing the current open-source ecosystem, ensuring that the benefits of technological innovation are truly accessible to all of humanity. This framework not only reflects China's active participation in global digital governance and its contribution to institutional public goods, but it also provides solid practical path for building a community of shared future for mankind.

Chapter Contents:

- 1.1 Open Source: A Public Good of the Digital Age and an Engine for Sustainable Development
- 1.2 OSDGs Framework: Four Pillars and Mapping to Sustainable Development



Four Pillars and Twelve Evaluation Dimensions

To truly reflect the decentralized, networked, and self-organizing nature of the open-source ecosystem, the OSDGs framework constructs a **Four Pillars Evaluation System (VCCE)**. This system is not a linear hierarchy but consists of four parallel, complementary dimensions that together form a multi-dimensional lens for measuring the health and social value of open source. Each pillar contains three specific evaluation dimensions, resulting in twelve core, measurable, and trackable indicators.

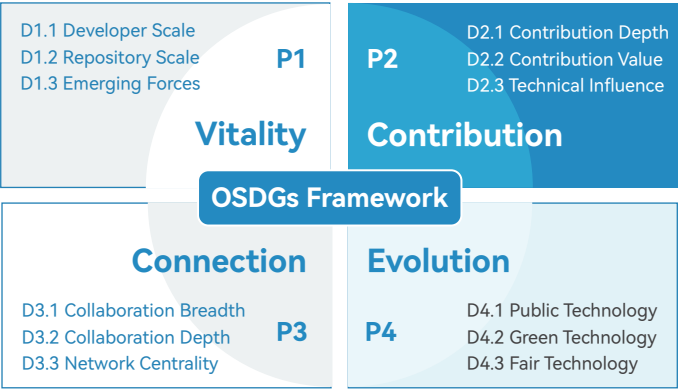


Figure 1.2.1 Overall framework of OSDGs

Table 1.2.2 Four Pillars of the OSDGs and Evaluation Dimensions

Pillar	English Name	Core Issue	Evaluation Dimensions	UN SDG Alignment
P1	Vitality	Who is involved? How active is the project?	D1.1 Developer Scale, D1.2 Repository Size, D1.3 New Blood	Inclusivity, Innovation, Sustainable Development
P2	Contribution	Is the contribution deep? Is collaboration valuable?	D2.1 Contribution Depth, D2.2 Contribution Value, D2.3 Technical Influence	Excellence, Impact, Knowledge Sharing
P3	Connection	How do developers collaborate? How are the world's ecosystems connected?	D3.1 Collaboration Breadth, D3.2 Collaboration Depth, D3.3 Network Centrality	Sustainable Cities, Inclusive Societies, International Cooperation
P4	Evolution	How can it grow sustainably? How does it serve the public good?	D4.1 Public Technology, D4.2 Green Technology, D4.3 Fair Technology	Clean Energy, Climate Action, Reduced Inequalities

Systematic Mapping with the United Nations SDGs

The OSDGs framework is not an isolated technical evaluation system but is deeply embedded in the global sustainable development agenda. Its four pillars and various SDGs are clearly mapped, providing value anchors for the open-source ecosystem.

Table 1.2.3 Mapping Relationship Between OSDGs and United Nations Sustainable Development Goals (SDGs)

OSDGs Pillars	Sub-Dimensions	Directly Supported SDGs	Support Logic
P1 Community Vitality	Developer Scale, Repository Size, New Blood	SDG 5 (Gender Equality), SDG 10 (Reduced Inequality)	Lowers participation barriers, promoting diverse groups' equal involvement in technology creation.
P2 Value Contribution	Technical Influence, Contribution Depth, Contribution Value	SDG 4 (Quality Education), SDG 17 (Global Partnerships)	Promotes knowledge sharing and builds cross-regional collaborative networks.
P3 Ecological Connection	Collaboration Breadth, Collaboration Depth, Network Centrality	SDG 11 (Sustainable Cities), SDG 16 (Inclusive Societies), SDG 17 (International Cooperation)	Enhances global collaboration, supporting smart cities and resilient community building.
P4 Sustainable Growth	Green Technology, Public Technology, Fair Technology	SDG 7 (Clean Energy), SDG 13 (Climate Action), SDG 10 (Reduced Inequalities)	Uses technology to drive environmental sustainability and social equity.

CHAPTER 2:

GLOBAL OPEN SOURCE OBSERVATORY

Chapter 2: Global Open Source Observatory

Open Source Pulse: Dynamic Map and Trend Insights of the Global Ecosystem

This chapter presents the **Global Open Source Observatory (GOSO)**, a dynamic, multidimensional map of the global open-source ecosystem. GOSO provides valuable insights through five key dimensions: activity, contribution, technology landscape, talent aggregation, and academic research. The system aims to deliver a comprehensive, visual representation of the ecosystem, showcasing geographic distribution, collaboration patterns, and development momentum, thus providing foundational, descriptive situational awareness for policymakers, businesses, and developers.

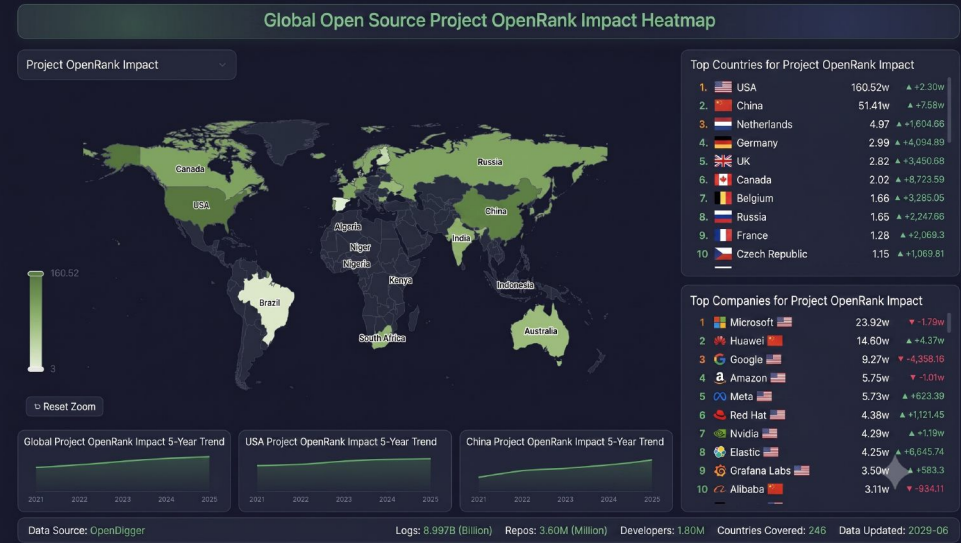
GOSO: Global Open Source Observatory

Chapter	English Name	Abbreviation	Content
2.1	Global Open Source Activity Map	A-Map	Reveals the overall scale of developer participation and geographic distribution, identifying the "geographical foundation" of the open-source ecosystem.
2.2	Global Open Source Contribution Map	C-Map	Focuses on tangible contributions such as code commits and issue resolutions, measuring the depth of technical input from different countries and regions.
2.3	Global Open Source Technology Map	T-Map	Displays trends and emerging hotspots in major technical areas, outlining the "technology landscape" of global innovation.
2.4	Global Open Source eXpertise Map	X-Map	Assesses the growth momentum and potential markets of developer communities in different countries, capturing sources of "new blood."
2.5	Global Open Source Research Map	R-Map	Tracks the production and dissemination of open-source related research, reflecting the academic evolution and theoretical development of the open-source ecosystem.

Key Country Rankings in Critical Indicators (Including China's Position)

Ranking	Active Developers	Active Growth	Open Source Contributions	Contribution Growth	Developer Influence	Influence Growth	Globalization	High-Impact Talent	Research Papers
1	USA	India	USA	China	USA	Netherlands	USA	USA	USA
2	China	China	China	Switzerland	China	Japan	UK	Germany	China
3	India	USA	Germany	Poland	Germany	France	Germany	China	Germany
4	Brazil	Brazil	UK	Netherlands	UK	Canada	India	UK	Brazil
5	Germany	Germany	France	India	India	UK	France	France	Canada
6	UK	Russia	Canada	France	France	Germany	Canada	Canada	France
7	Canada	Canada	India	Canada	Canada	Brazil	Netherlands	Netherlands	UK
8	France	UK	Poland	UK	Brazil	India	Japan	Australia	Australia
9	Russia	Indonesia	Netherlands	Germany	Netherlands	China	Brazil	Australia	Netherlands
10	Poland	France	Switzerland	USA	Japan	USA	China	Japan	India
China's Position	2	2	2	1	2	9	10	3	2





CHAPTER 3:

GLOBAL OPEN SOURCE INFLUENCE RANKING

Global Open Source Influence Ranking

Section	English Title	Abbreviation	Definition	Typical Scenarios
3.1	Global Region Influence Ranking	R-Ranking	Reveals the overall strength and strategic layout of open source ecosystems at the national and regional levels from a geographical perspective.	Government science and technology policy departments, international organizations, regional innovation centers
3.2	Global Corporate Influence Ranking	C-Ranking	Demonstrates the leadership and contribution depth of tech giants and open source-native enterprises within the ecosystem from an organizational perspective.	Corporate strategy divisions, investment institutions, technology M&A teams
3.3	Global Technology Influence Ranking	T-Ranking	Identifies high-potential technological tracks and trends — such as Artificial Intelligence, Cloud Computing, and RISC-V — from a directional perspective.	CTOs, system architects, technology selection committees
3.4	Global Project Influence Ranking	P-Ranking	Focuses on the most influential mature open source projects — such as OpenHarmony and PyTorch — from an outcome-oriented perspective.	Developers, project maintainers, foundations, media organizations
3.5	Global Emerging Project Influence Ranking	EP-Ranking	Captures emerging projects with explosive growth potential, anticipating next-generation innovation hotspots from a future-oriented perspective.	Venture capital firms, incubators, innovation labs, university research teams

2026 Global Open Source Development Composite Ranking (Top 10 Countries)

Rank	Active Developers	Active Growth	Contributions	Contribution Growth	Developer Influence	Influence Growth	Degree of Globalization	Research Papers	Regional Ranking	Corporate Ranking	Project Ranking	Talent Ranking
1	United States	India	United States	China	United States	Netherlands	United States	United States	United States	United States	United States	United States
2	China	China	China	Switzerland	China	Japan	United Kingdom	Germany	France	China	China	Germany
3	India	United States	Germany	Poland	Germany	France	Kingdom of Germany	China	Germany	Germany	Germany	China
4	Brazil	Brazil	United Kingdom	Netherlands	United Kingdom	Canada	India	United Kingdom	China	France	Netherlands	United Kingdom
5	Germany	Russia	France	India	India	United Kingdom	France	Italy	India	Canada	Russia	France
6	United Kingdom	Indonesia	Canada	France	France	Germany	Canada	Canada	Canada	Sweden	Belgium	Canada
7	Canada	Canada	Netherlands	United Kingdom	Brazil	Brazil	Netherlands	Spain	Switzerland	Estonia	Canada	Netherlands
8	France	United Kingdom	Netherlands	United Kingdom	India	India	Japan	France	Poland	Bulgaria	South Africa	Switzerland
9	Russia	Indonesia	Poland	Germany	Netherlands	China	Brazil	Brazil	Australia	United Kingdom	Iceland	Australia
10	Poland	France	Switzerland	United States	Japan	United States	China	Netherlands	Brazil	Norway	Norway	Japan
China's Position	2	2	2	1	2	9	10	2	4	2	2	3

Global Domain Project OpenRank Leaderboard				
Filter: Global, Projects Period: 2026-05 Records: 300 Data Updated: 2026-06				
#	Project Name		OpenRank	Participant Count
1		OpenHarmony OpenHarmony aims to be a next-generation distributed operating system for all scenarios, connect all devices, and form a platform for smart endpoint terminals.	6050.4 ↓ 69.3	2823 ↓ 139
2		CANN CANN (Compute Architecture for Neural Networks) is Huawei's core chip-level acceleration architecture designed for AI scenarios.	2949.1 ↑ 434.8	2502 ↑ 329
3		Ascend Fully supports Ascend AI processors for application development, training acceleration, cluster management, and key open-source software tools.	1785.5 ↑ 205.1	1503 ↑ 239
4		openEuler openEuler is building an innovative community and platform, creating a unified and open operating system that supports multiple processing architectures.	1537.3 ↑ 93.0	1027 ↓ 22
5		Azure Microsoft Azure API, SDK, and other open-source projects.	1498.1 ↓ 277.6	932 ↓ 297
6		LLVM LLVM is a collection of modular compiler and toolchain technologies for building highly optimized compilers and runtimes.	1323.8 ↓ 182.2	764 ↓ 248
7		OpenClaw A personal AI assistant for any operating system and platform.	1068.4 ↓ 300.0	1485 ↓ 1532
8		DataDog Cloud application observability and security platform.	1027.3 ↓ 195.4	458 ↓ 162
9		NixOS Nix package management and purely functional Linux distribution.	1023.4 ↓ 136.1	970 ↓ 230
10		OpenShift A developer and operations friendly Kubernetes distribution.	1005.0 ↓ 247.1	595 ↓ 226



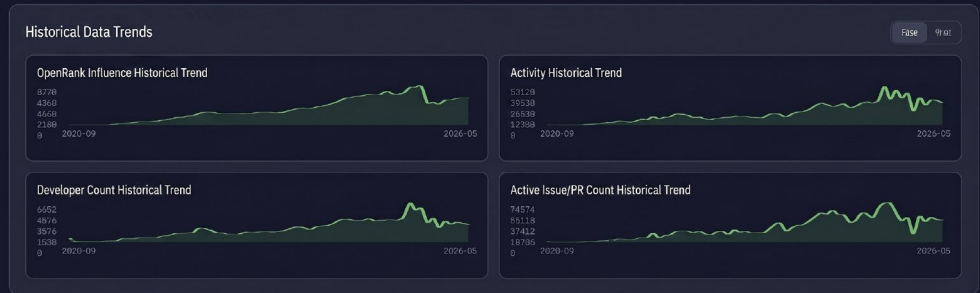
OpenHarmony
OpenHarmony aims for full-scenario, full-connection, full-intelligence, building a smart terminal operating system framework and platform.
Repository
[openharmony](#) [openharmony](#) [openharmony-sig](#) [openharmony-tpc](#) [openharmony-sig](#)

Contribute

**6,050.35**
OpenRank Influence (2026-05)
▲ 69.3
▲ 1.1%

**30,552.41**
Activity (2026-05)
▲ 3,021
▲ 9%

**2,823**
Developers (2026-05)
▲ 139
▲ 4.7%



Community Developer Contributions				
2026-04 1 / 10				
Rank	Platform	Name	Contributions Dashboard	OpenRank Change
1		lyong_sd		52.4 ▲ 5.3
2		agentic-workflow		46.3 -
3		cheil		34.5 ▼ 9.2
4		qp_4088541		33.6 -
5		yeyuming		32.2 -

OpenRank Global Initiator Leaderboard

Current View: Global, Initiator, 2026.05 | Results: 300 | Data Updated: 2026-06

#	Initiator Name	OpenRank Score	Participant Count
1	Huawei	13639.6 ↑ 680.5	8448 ↑ 498
2	Microsoft	7268.9 ↓ 1375.1	4770 ↓ 1628
3	Google	2935.5 ↓ 662.9	2503 ↓ 1090
4	NVIDIA	2453.9 ↑ 195.0	1284 ↓ 375
5	Elastic	2266.3 ↓ 304.9	621 ↓ 113
6	RedHat	1801.5 ↓ 423.1	1052 ↓ 392
7	Meta	1718.7 ↓ 401.7	1092 ↓ 456
8	Amazon	1462.8 ↓ 472.5	1023 ↓ 522
9	Anthropic	1339.8 ↓ 589.4	2184 ↓ 4420
10	University of Illinois Urbana-Champaign	1323.8 ↓ 182.2	764 ↓ 248



Microsoft

Company

Repository List

microsoft

Azure

dotnet

PowerShell

dapr

Inaugural Year

7,268.86

OpenRank Influence (2026-05)

1,375.1
15.9%

12,531.23

Activity Score (2026-05)

7,432.6
37.3%

4,770

Developers (2026-05)

1,828
27.7%



CHAPTER 4:

GLOBAL OPEN SOURCE TECHNOLOGY LANDSCAPE INSIGHT

Chapter 4: Global Open Source Technology Landscape Insight

Global Open Source Technology Landscape Insight: From Foundational Infrastructure to Frontier Applications

This chapter presents the Global Open Source Technology Landscape Insight (GOSL), which aims to systematically map the key domains and evolving trends within the global open source technology ecosystem. As core technologies such as artificial intelligence, cloud computing, big data, and databases increasingly converge, open source has emerged as a central engine driving the evolution of digital infrastructure.

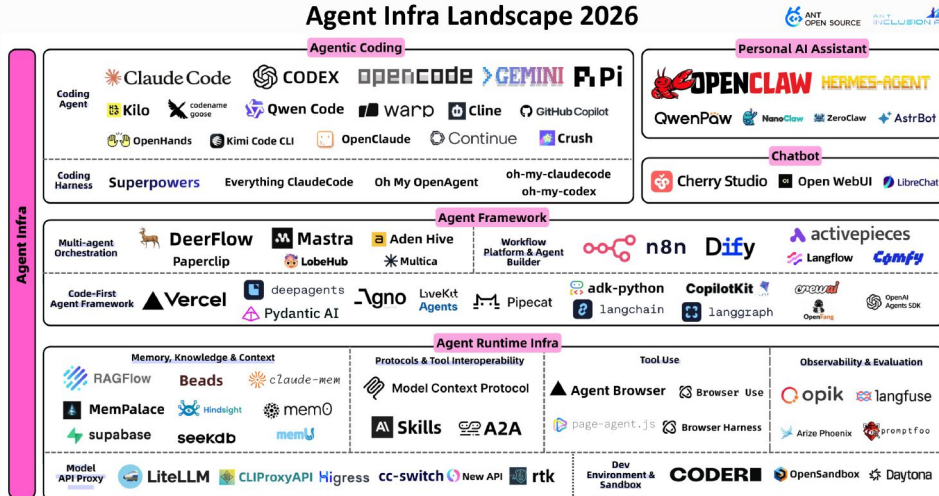
The GOSL framework adopts the **OpenRank** influence index as its core evaluation standard, integrating multi-source datasets—including GitHub collaboration behaviors, DB-Engines market dynamics, and the CNCF technology landscape—to construct a three-tier quantitative analytical model built around Behavior–Quality–Influence. This enables dynamic, comparable, and cross-layer assessment of the open source technology ecosystem.

Based on a comprehensive selection and analysis of open source projects, this chapter focuses on four core domains and one international foundation: **Artificial Intelligence and Large Models (A)**, **Big Data and Data Engineering (B)**, **Cloud Infrastructure (C)**, **Database Systems (D)**, and the **PyTorch Foundation (P)**. Covering perspectives from foundational infrastructure to upper-layer applications, the GOSL employs multidimensional data analysis and case studies to illuminate the technological evolution pathways and future directions across these key areas.

GOSL: Global Open Source Technology Landscape Insight

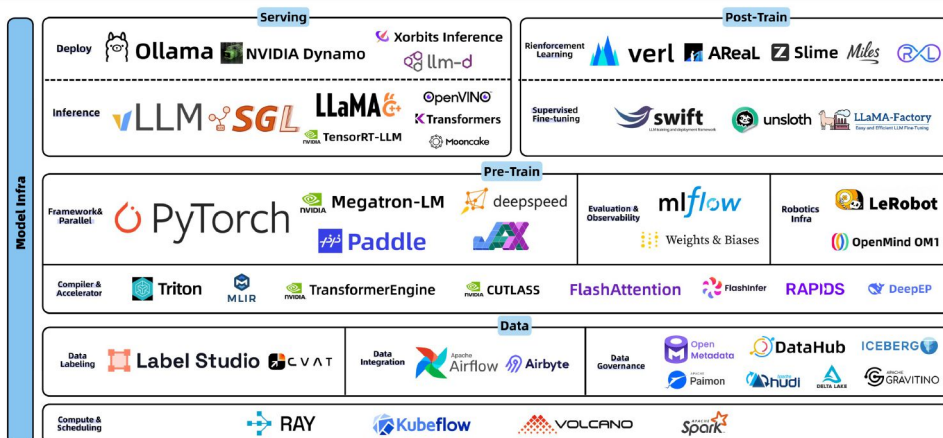
Chapter	Technical field	Abbreviation	Number of selected projects
4.1	AI and Large Model Landscape Insight	A-Landscape	114
4.2	Big Data and Data Engineering Landscape Insight	B-Landscape	97
4.3	Cloud Infrastructure Landscape Insight	C-Landscape	97
4.4	Database Systems Landscape Insight	D-Landscape	91
4.5	PyTorch Foundation Landscape Insight	P-Landscape	61

Agent Infra Landscape 2026



<https://github.com/antgroup/agent-infra-landscape>

Model Infra Landscape 2026



<https://github.com/antgroup/agent-infra-landscape>

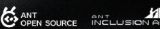
Agents and Infrastructure Are Becoming Co-Centers

#	Projects	OpenRank	Participants	Trend	Introduction
1	OPENCLAW	1,368.43	2,980		Personal Agent / coding / workflow
2	vLLM	1078.67	1,283		LLM Inference
3	PyTorch	1041.30	620		Deep Learning Infrastructure
4	Claude Code	810.36	4,015		Coding Agent
5	SGL	497.96	530		LLM Inference
6	CODEX	474.90	1,069		Coding Agent
7	HERMES-AGENT	473.32	1,651		Personal / autonomous agent
8	openCODE	471.04	1,124		Open source compatible coding agent
9	LangChain	359.67	429		Agent Framework
10	GEMINI	355.52	474		Coding Agent

Sources: GitHub & OpenDigger

Period: April 2026

Coding Agent: Silicon Execution Layer



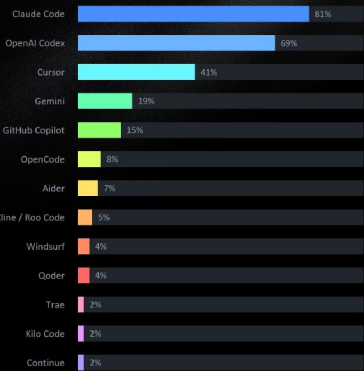
OpenRank Top 100 Agentic AI Projects · 2026-05



Projects using the most agents

1	google/adm-python	8
2	CherryHQ/cherry-studio	6
3	BerriAI/litellm	5
4	langgenius/dify	5

Adoption by Coding Agent



Method: scanned repository file trees via GitHub API, identifying marker files such as CLAUDE.md / cursor / AGENTS.md / codex / windsurfrules / gllgnore.

CHAPTER 5:

STATE OF OPEN SOURCE AI

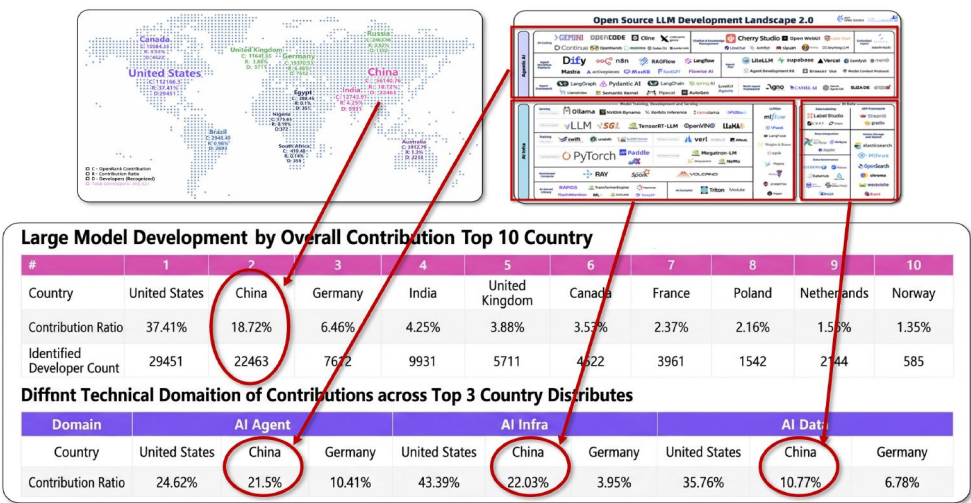
Chapter 5: State of Open Source AI

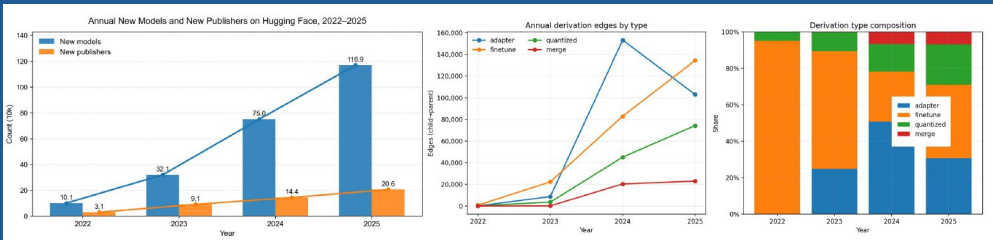
Open Source Artificial Intelligence: From Conceptual Consensus to Institutional Framework Construction

This chapter presents the "Current Development Status of Open Source Artificial Intelligence" (Open Source AI, OSAI), systematically showcasing the evolution of the global large-model open-source ecosystem from conceptual advocacy, technological practices, to institutional development. As AI large models become the core area of global technological competition, open source has evolved from a mode of technological collaboration to a key mechanism for promoting inclusive innovation, breaking technological monopolies, and achieving global co-governance. This chapter, based on academic research, industry patents, and open-source ecosystem dimensions, reveals China's leadership in scale and application, the U.S.'s continued advantage in frontier exploration and influence, as well as the global institutional response to the "true open-source" standard.

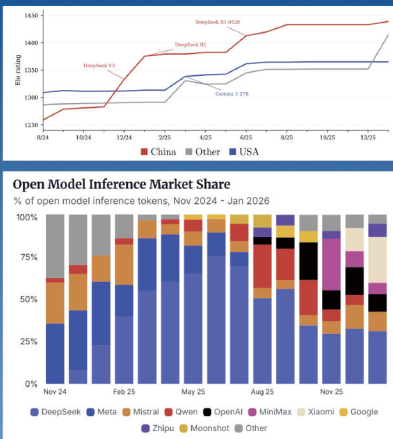
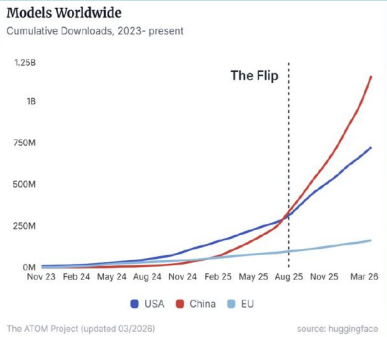
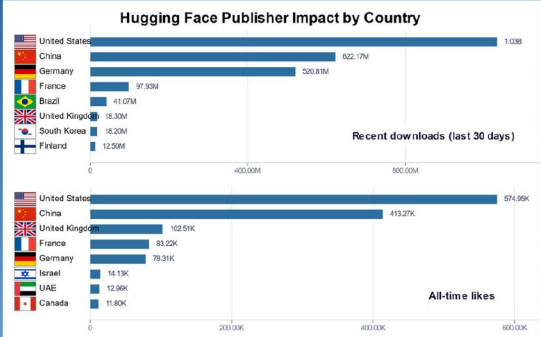
OSAI focuses on three core sections: First, **the deepening of the open-source concept and local practices** demonstrates that domestic open-source large models represented by Qwen and DeepSeek are driving the construction of an ecosystem integrating "accessibility, publicness, inclusivity, and security." Developers widely recognize the value of open source in reducing costs, improving efficiency, and adapting to various scenarios, though concerns about "insufficient openness" and "unknown security risks" remain prominent. Second, **the institutionalization of openness** responds to the crisis of trust by proposing an open-source grading system centered on "capacity level × usage license," and advocates for the establishment of a global unified "AI large model open-source label," pushing platforms like Hugging Face and ModelScope to automatically tag content, ensuring that "understandable, modifiable, verifiable, and co-constructable" truly defines open-source. Finally, **quantitative insights into the global innovation landscape** show that China leads in total AI papers and patents, while the U.S. maintains an advantage in highly cited papers and academic influence, forming a "China-U.S. dual-core, multi-country advancing" competitive collaboration pattern.

The following table shows the ranking of major countries in terms of AI large model patent papers:





Open Source models from China are set to over take the US, as downloads surge in usage across HuggingFace. The American ecosystem has stalled in growth, and the “flip” is happening right now.



The **DeepSeek** series is driving Chinese large models to achieve a comprehensive leap—from performance leadership to ecosystem dominance.

China's open-source LLM cluster dominates global inference: **DeepSeek, Qwen, Moonshot, Z.ai, MiniMax, and Xiaomi** collectively command over **70%** of the market, propelling open models from “technically viable” to “industrial mainstream.”

China

Nearly 80% of new AI startups are now building on Chinese open-source infrastructure. (Andreessen Horowitz (a16z))

Open-Source and Openness Level Assessment of Large Language Models



Organization-Type View



CONCLUSION:

TOWARDS A SUSTAINABLE DIGITAL FUTURE

Conclusion: Towards a Sustainable Digital Future

In an era where digital technologies are deeply embedded in the fabric of human society, open source has transcended its origins as a mere software development methodology to emerge as critical infrastructure underpinning global innovation, social equity, and sustainable development. This report provides a comprehensive analysis of the current state, key trends, and strategic value of the global open source ecosystem, offering policymakers, technologists, research institutions, and international organizations a data-informed, practice-oriented, and values-driven perspective that integrates technical, governance, and ethical dimensions.

The report affirms that open source is increasingly recognized by the international community as a new form of global digital public good. Its non-excludable, non-rivalrous nature and strong positive externalities enable it to transcend national borders, industrial sectors, and socio-economic divides, serving as a foundational platform for knowledge sharing, collaborative innovation, and equitable access to capabilities. The 17 Sustainable Development Goals (SDGs) enshrined in the United Nations 2030 Agenda—ranging from poverty eradication and quality education to climate action and peaceful, just societies—are increasingly supported by open source solutions. Whether through open-source software lowering barriers to technology adoption, open data enhancing transparency in public services, or open artificial intelligence empowering marginalized communities, open source is contributing in concrete and measurable ways to the realization of the SDGs.

To operationalize this contribution, the report introduces the **Open Source Development Goals (OSDGs)** analytical framework, which translates the broad vision of the SDGs into actionable technical pathways and assessment criteria. From an **environmental (E)** standpoint, open source promotes green and low-carbon transitions by optimizing resource efficiency, improving energy performance, and enabling climate modeling. From a **social (S)** perspective, it advances educational equity, strengthens digital inclusion, and expands technological access for vulnerable populations. From a **governance (G)** angle, open source enhances system transparency, bolsters the security and resilience of critical digital infrastructure, and offers practical models for multi-stakeholder co-governance.

Within the evolving global technology landscape, open source has become a pivotal force in democratizing innovation. In particular, open large language models—such as Qwen, Llama, and DeepSeek — are dismantling technological monopolies and accelerating the shift of AI from exclusive laboratories to widespread, real-world applications across industries. The report proposes a **Tiered Openness Assessment Framework for Open Foundation Models**, establishing quantifiable and verifiable technical standards to define “true openness.” This initiative responds to urgent industry demands for transparency and trustworthiness and contributes a normative instrument to global AI governance.

At the same time, the report acknowledges persistent challenges: rising techno-nationalism, geopolitical fragmentation of open source ecosystems, insufficient mechanisms for the sustainable maintenance of critical digital infrastructure, and governance gaps concerning security, ethics, and accountability in open source projects. These issues underscore a fundamental truth: the potential of open source does not automatically translate into public value. Its long-term sustainability requires deliberate institutional support, sustained investment, and strengthened international cooperation.

In this context, the report puts forward the **Global Digital Public Goods Coding System (GDPG-CS)** — a

unified identification and classification framework for open source projects, open datasets, AI models, and technical standards. Beyond facilitating discovery and impact assessment, this system aims to serve as a bridge between technical practice and policy formulation, enabling a transition from fragmented initiatives toward systemic, coordinated governance.

At its core, open source is about collaboration; its future lies in shared stewardship. In the face of global challenges, no single nation, corporation, or community can act alone. Only through steadfast commitment to the principles of openness, transparency, and inclusivity—and through the active participation of governments, the private sector, academia, and civil society—can we build a more equitable, trustworthy, and sustainable digital civilization.

We affirm that open source innovation belongs to all humanity, and our digital future must be co-created. With continued investment, institutional innovation, and global solidarity, open source will remain a powerful engine for scientific advancement and social progress, providing robust support for the achievement of the United Nations Sustainable Development Goals and the realization of a shared

APPENDIX A:

INDICATOR SYSTEM AND DATA SOURCES

Appendix A: Indicator System and Data Sources

This report constructs a global open-source development assessment system based on the series of standards for open-source governance in information technology issued by the Ministry of Industry and Information Technology of the People's Republic of China. It integrates the United Nations Sustainable Development Goals (SDGs) and Open Source Development Goals (OSDGs) concepts, and innovatively proposes a progressive three-layer evaluation framework of “Behavior—Quality—Impact”. This indicator system forms the core methodology for **Chapter 2, "Global Open Source Observatory" (GOSO)**, **Chapter 3, "Global Open Source Influence Ranking" (GOSR)**, and **Chapter 4, "Global Open Source Technology Landscape Insight" (GOSL)**, ensuring the scientific, comparable, and traceable nature of the entire process—from ecosystem observation to influence ranking and technology trend analysis.

1、Source and Theoretical Foundation of the Indicator System

The indicator system used in this report is primarily based on China’s first systematic and structured open-source governance standard framework—the “Information Technology Open Source Governance” series of national standards (GB/T), specifically including:

- **Part 3: Community Governance Framework:** Defines community developer activity, influence, and network value indicators;
- **Part 4: Project Evaluation Model:** Defines project health, influence, and maturity indicators;
- **Part 5: Open Source Contributor Evaluation Model:** Defines statistical and network evaluation methods for measuring developer contributions.

Building on this foundation, this report integrates the **Open Source Development Goals (OSDGs)** framework to construct a comprehensive evaluation system that covers the “Behavior—Quality—Impact” three-tier progressive logic. The system aims to scientifically, fairly, and comparably assess the development trend of the global open-source ecosystem, contributing to the construction of digital public goods and global technology governance.

2、Definition of Core Contribution Behaviors

To comprehensively reflect the full process of open-source collaboration, this report adopts the following five categories of community contribution behaviors as the fundamental data units. These behaviors cover the entire chain, from “proposing ideas” to “code implementation,” embodying the core concept that “collaboration is contribution.”

Abbreviation	Name	Definition and Explanation
OI	Open Issue	Raising new issues, proposing feature suggestions, or engaging in technical discussions is the first step toward participating in open source.
ID	Issue Discussion	Engage in existing issues or PRs by contributing information, answering questions, or helping others.
OPR	Open Pull Request	Submitting a pull request marks the beginning of tangible code contributions.
PRR	Pull Request Review	Reviewing others’ code, offering revision suggestions, or approving changes is a critical step in ensuring code quality.
MPR	Merge Pull Request	Once the code passes review and is officially merged into the main branch, it signifies the completion of a full collaboration cycle.

Simplified Understanding:

- OI: “I have an idea, can we discuss it?”
- ID: “I’ll join the discussion and offer some suggestions.”
- OPR: “I’ve written the code, can it be merged?”
- PRR: “I’ll help review the code to see if there are any issues.”
- MPR: “Approved! The code is officially added to the project!”

3、Indicator Dimension System

This report constructs a four-dimensional progressive indicator system—“**Activity—Contribution Volume—Contribution Degree—Impact**”—which corresponds to the four levels of “whether there is participation,” “how much is contributed,” “the quality of participation,” and “the scope of impact”, forming a complete evaluation chain from micro-level behaviors to macro-level ecosystem dynamics.

(1) Activity

Measures the breadth of participation by a community or individual, answering the question, “Is anyone using it?”

Concepts and Metrics	Terms and Abbreviations	Definition	Application Scenarios
Active developer	Active Contributor (AC)	Developers who have performed at least one OI/ID/OPR/PRR/MPR action within the statistical period.	Identifying genuine participants
The number of active developers	Number of Active Contributors (NAC)	Total number of developers who meet the definition of an active developer.	Measuring talent scale and ecosystem attractiveness.
Active open source repository	Active Open Source Repository (AOSR)	Repositories in which at least one of the five types of actions occurred during the statistical period.	Excluding ‘zombie repositories’ to focus on genuinely active ones.
The number of active open-source repositories	(Number of Active Open Source Repositories, NAOSR)	The total number of active repositories on a given code hosting platform.	Evaluating the overall level of activity.
Active open source projects	Active Open Source Project (AOSP)	Projects in which at least one of the five types of actions occurred during the statistical period.	Excluding ‘zombie projects’ to focus on genuinely active ones.
The number of active open-source projects	Number of Active Open Source Projects (NAOSP)	The total number of active projects within a specific scope (country, enterprise, or technical domain).	Evaluating the overall health of the ecosystem.

(2) Contribution Volume

This measures the frequency of behaviors in statistical terms, reflecting the intensity of participation, and is suitable for macro-level trend analysis.

- **Behavior Counting Rules:** Each type of behavior is counted independently (e.g., each comment or PR submission is counted as one instance).
- **Individual/Organization Contribution Volume:** The cumulative count of all five behaviors forms the “contribution profile.”
- **Project Contribution Total:** The total sum of all behaviors received by the project, reflecting the intensity of collaboration.

Note: Contribution volume does not differentiate behavior quality and is primarily used for trend monitoring and preliminary screening.

(3) OpenRank Contribution Impact

This indicator, calculated based on the “Community OpenRank Algorithm,” is used to evaluate the quality and depth of developer contributions within a project. It is suitable for organizations and individual developers.

- **Calculation Logic:** It constructs a heterogeneous collaboration network within a project, considering factors such as behavior type, adoption rate, and reviewer weight, to calculate the relative contribution value of a developer in that community.
- **Application Scenarios:** These include identifying high-quality core contributors, supporting community incentive mechanisms, and assessing the actual value of employees in key projects.

(4) OpenRank (OpenRank Influence)

This indicator, calculated based on the “Global OpenRank Algorithm,” is used to assess the network position and systemic impact of entities within the global open-source ecosystem. It is suitable for multi-level entities:

Entity Types	Indicator Name	core value
Individual / Group	Individual OpenRank Influence	Identifying cross-project technical leaders and ecosystem connectors.
Project / Repository	Project OpenRank Influence	Evaluating the degree of dependency and leadership that a project holds within the global ecosystem.
Enterprise / Organization	Enterprise OpenRank Influence	Measuring an enterprise’s actual influence in open-source governance and standards development.
Country / Region	Country OpenRank Influence	Measuring an enterprise’s actual influence in open-source governance and standards development.
Technology Ecosystem	Technology Ecosystem OpenRank Influence	Assessing the overall development momentum of a specific technological field (e.g., AI or cloud-native).

4、Open Source Development Matrix (OSDM)

To balance the needs of macro-level analysis and ranking assessments, this report constructs the **Open Source Development Matrix (OSDM)**. This matrix organizes indicators in a hierarchical, dimensional, and scenario-based manner, supporting the ecosystem observation in **Chapter 2 (GOSO)** and the influence ranking in **Chapter 3 (GOSR)**.

Level / Dimension	Activity (Macro)	Contribution Volume (Macro)	Contribution Degree (Macro)	Influence (Ranking)
Individual	Number of active developers	Total Individual Contributions	High-Impact Developers Ratio	Aggregated Individual Influence
Project	Number of active projects	Total Project Contributions	Core Projects Ratio	Aggregated Project Influence
Enterprise (from a People Perspective)	The number of developers within the enterprise	—	—	Enterprise (Organizational) Influence
Enterprise (from a Project Perspective)	The number of active projects associated with the enterprise	—	—	Associated Project Ecosystem Influence
Technology	The number of active projects in the technical field	—	—	Technology Ecosystem Influence
Country & Region	The total number of active national developers	Total National Contributions	High-Impact Developers Ratio	National Influence (Ranking)

5、Data Sources and Calculation Notes

This report uses a unified 12-month rolling statistical period. The data period for 2025 is from **July 1, 2024, to June 30, 2025**, with subsequent years following this same cycle to ensure the comparability and consistency of cross-year trend analysis.

The data collection and analysis for this study rely on the **OpenDigger open-source project** (<https://github.com/X-lab2017/open-digger>) and its corresponding **online service platform** (<https://open-digger.cn>). All behavioral data is systematically collected and processed through this platform.

OpenDigger is an open data infrastructure aimed at promoting transparency and measurability in the global open-source ecosystem. Initially launched by the X-lab Open Laboratory, it grew within the Mulan Open Source Community and integrates research outcomes from universities, research institutions, and enterprises. To advance the transparency, measurability, and sustainable development of the open-source ecosystem, OpenDigger is building an open, neutral, and multi-stakeholder global governance framework, striving to become a public digital infrastructure that serves global open-source governance.

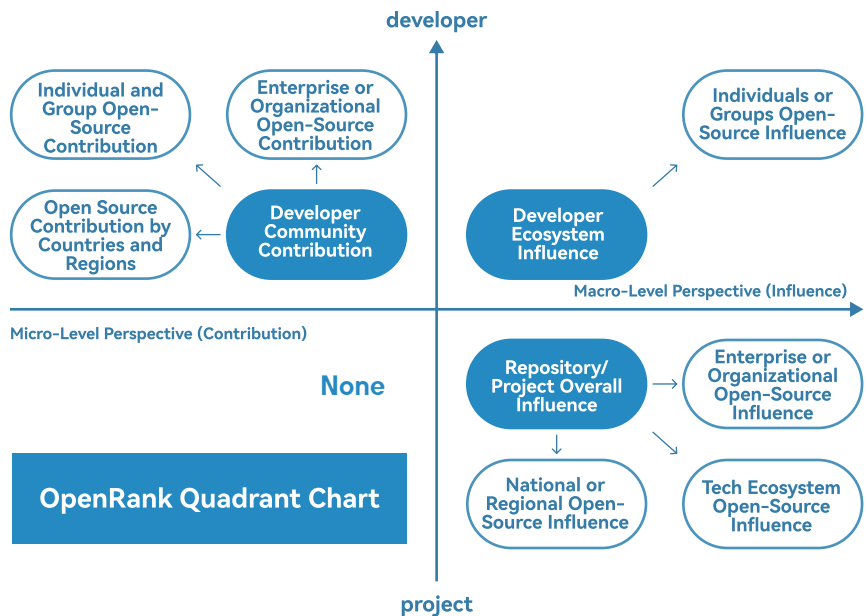
Currently, the project is led by the United Nations Advisory Committee on Open-Source Innovation (UNOSI) and is working toward operational sustainability. It has established collaborations with

international organizations such as the United Nations International Computing Centre (UNICC), the BenchCouncil International Open Benchmarking Committee, and the Linux Foundation, exploring open-source data standards, technical evaluation systems, and global collaboration mechanisms.

As of now, OpenDigger has collected over **8 billion** collaboration behavior logs, **6.2 million** open-source artifact repository data, **160,000** CVE security vulnerability records, and **25 million** StackOverflow technical Q&A entries, among others. This has created a comprehensive, high-quality, and continuously updated open-source big data set, providing strong data support for community governance analysis, developer evaluation, and project influence assessment in this report.

In this report, the core evaluation metric—**OpenRank North Star Metric**—is designed based on OpenDigger’s multi-dimensional data system. OpenRank is the world’s first systematic and original open-source standard framework, and it has been included by the Ministry of Industry and Information Technology in the “2024 Group Standards Application and Promotion Typical Cases” (MIIT Science and Technology Letter [2025] No. 19), reflecting its authority and practical value.

As shown in the figure, the OpenRank indicator system is built from a “**Developer-Project**” two-dimensional perspective, forming a quadrant model that includes the **OpenRank Contribution Impact** and **OpenRank Influence** dimensions. It comprehensively characterizes the quality of individual and project participation in open-source and their social value.



At the same time, **OpenRank**, as one of the core outputs of the OpenDigger platform, possesses the following five key characteristics:

- **Global Perspective:** It comprehensively considers a developer’s behavioral trajectory across multiple platforms and projects.

- **Focus on Influence:** It emphasizes not only the quantity of contributions but also the spread of technical knowledge and the position within collaboration networks.
- **Encouragement of Collaborative Interaction:** It values interactive behaviors such as Issues, Pull Requests, and comments.
- **Encouragement of Long-Term Engagement:** It introduces a time decay mechanism to incentivize sustained contributions.
- **Robustness and High Scalability:** It supports unified evaluation across platforms, languages, and domains.

For detailed documentation, please refer to: <https://open-digger.cn/docs/user-docs/intro>.

All data used in the 2026 Global Open Source Development Report has been made publicly available in conjunction with the OpenDigger project, ensuring transparency and reproducibility of the research. For data details, please refer to:

<https://github.com/X-lab2017/open-digger/tree/master/notebook/GOSD2026>.

Open Source Development Index (OSIDX)

Tier 1 (Weight)	Tier 2	Tier 3 (Measurable Metrics)
1. Ecosystem Foundation (25%)	1.1 Developer Scale	• Active contributors (NAC) • % High-frequency contributors (≥ 10 actions/month)
	1.2 Project Activity	• Active repositories (NAOSR) • Avg. monthly activity per repo (commits + issues + PRs)
	1.3 Growth Momentum	• YoY growth in new active contributors • % First-time or newcomer contributors
2. Contribution Quality (35%)	2.1 Behavioral Depth	• % Code reviews (PRR) • % Contributions to core branches (e.g., main)
	2.2 Value Recognition	• PR merge rate (MPR / OPR) • % High-impact PRs (used by ≥ 3 projects)
	2.3 Network Influence	• OpenRank Influence Score • Number of direct + indirect dependent projects
3. Global Collaboration (25%)	3.1 Cross-Border Reach	• % Contributions to foreign-hosted projects • Number of collaborating countries (≥ 5 major regions)
	3.2 Multilateral Engagement	• Number of projects in global foundations (e.g., CNCF, Apache) • Number of led standards or specification proposals
	3.3 Inclusivity	• % Repositories with multilingual support • Proxy indicators for underrepresented group participation
4. Sustainable Innovation (15%)	4.1 Academic Output	• Annual count of top-tier CS papers on open source • % Papers with public code links
	4.2 Talent Pipeline	• Number of high-influence developers (OpenRank ≥ 200) • 3-year contributor retention rate
	4.3 Public Value	• Number of SDG-aligned projects (e.g., climate, health, education) • Usage volume of open datasets or tools

Acknowledgements

This report was guided by the China Association for Science and Technology (CAST), leveraging its consultative status with the United Nations Economic and Social Council (ECOSOC). It was supported and co-led by the CAST Advisory Committee on Open Source Innovation (CCOSI), with the Singularity Intelligence Research Institute serving as the publishing entity. The Chinese Institute of Communications (CIC) served as the secretariat of the committee, responsible for organizational coordination and implementation support.

The report is the result of collective wisdom and substantive contributions from experts across the global technology, academic, industrial, and international communities, reflecting the open, collaborative, and co-governed spirit of open source.

In terms of policy framework and strategic direction, UNESCO, the International Telecommunication Union (ITU), the United Nations International Computing Centre (UNICC), and the Office of the UN Secretary-General's Envoy on Technology (OSEDET) provided not only policy references but also critical input through expert engagement. Their insights on core themes, analytical frameworks, and alignment with global development initiatives ensured the report's deep alignment with the 2030 Agenda for Sustainable Development.

Scholars and experts from leading universities and research institutions—including Tsinghua University, Peking University, Shanghai Jiao Tong University, Fudan University, East China Normal University, University of Chinese Academy of Sciences, Nanjing University, Beihang University, Beijing University of Posts and Telecommunications, National University of Defense Technology, The University of Hong Kong, University of International Business and Economics, Beijing Institute of Technology, Northeastern University, the Strategic Consulting Institute of the Chinese Academy of Sciences, the Documentation and Information Center of the Chinese Academy of Sciences, Shanghai Artificial Intelligence Laboratory, and Alibaba Research—deeply participated in theoretical development, data analysis, and chapter drafting, providing robust support in foundational research, trend analysis, and science and technology policy.

Additionally, scholars from internationally renowned institutions such as the National University of Singapore, the University of Oxford, the University of California, Berkeley, Case Western Reserve University, and the Irish Software Research Centre contributed cutting-edge research and cross-national comparative perspectives in the fields of open source governance, innovation ecosystems, and the societal impact of technology, enriching the report's global analytical dimensions.

The China Electronics Standardization Institute (CESI) and the China Academy of Information and Communications Technology (CAICT), as national key institutions in information technology and standardization, provided professional support in technical standard referencing, open source evaluation system design, policy compliance review, and data methodology development, significantly enhancing the report's technical rigor and policy relevance.

On the industrial and ecosystem front, engineers and open source governance experts from Huawei OSPO, Alibaba OSPO, Ant Group OSPO, ByteDance OSPO, Baidu OSPO, and other enterprises shared full-stack technical practices and global ecosystem-building experiences. Representatives from international open source foundations—including LF AI & Data, LF OpenInfra, and LF PyTorch—assisted in refining comparative analyses of open governance models. Members of community organizations such

as the China Open Source Promotion Union (COPU), KAIYUAN.SHE, X-lab Open Source Research Institute, ALC-Beijing, and ModelScope Community contributed voluntarily to content discussions and case collection. CSDN, as a developer platform, supported broad technical trend surveys and the dissemination of open source concepts.

The Benchmark Council provided methodological guidance and data standardization support for the report's technical evaluation framework, ensuring the scientific validity, transparency, and comparability of the research findings.

This report is a product of multi-stakeholder collaboration, with every section shaped by the insights of experts across diverse fields. We extend our sincere gratitude to all contributing institutions and individuals. Moving forward, we will continue to uphold an open collaboration mechanism and welcome more organizations and individuals committed to open source innovation and global digital equity to join future research efforts, collectively advancing open source as a key enabler of the Sustainable Development Goals.

In the world of open source Every contribution deserves its reward

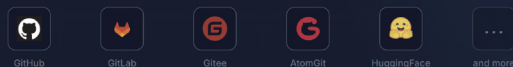
OpenShare connects cross-platform contributions, developer capability, organization demand, and points incentives into a verifiable value protocol for

Open Source Value Network

-  **Integrated Platforms** 5+
Normalize open source signals from GitHub, Gitee, GitLab, and more
-  **Developer Profiles** 28M+
Capture capability, collaboration quality, and growth trajectory
-  **Project Analysis** 48M+
Read project activity, technical influence, and ecosystem position
-  **Points Allocated** 18M+
Turn contribution recognition into durable incentive flow

DATA SOURCES

Aggregating Global Open Source Ecosystem Data



Building a Complete Open Source Talent Ecosystem

Deep Open Source Talent Insights Insight

Build developer capability profiles and value assessment systems based on multi-dimensional open source contribution data. Quantify technical capabilities, collaboration quality, and contribution trends to provide data support for talent decisions.

- Multi-dimensional Contribution Assessment
- Technical Domain Proficiency Analysis
- Time-series Capability Tracking
- Developer Ranking & Comparison

Precise Talent Outreach Talent Reach

An intelligent filtering and targeted outreach platform based on developer profiles. Help enterprises and communities efficiently connect with target talents and achieve precise matching of talent supply and demand.

- Smart Multi-dimensional Filtering
- Targeted In-platform Outreach
- Performance Data Tracking
- Developer Incentive Mechanism

Contribution Incentive System Credits

Build an ecosystem for open source contribution value circulation. Enterprises can grant incentives to core contributors, developers receive continuous recognition and rewards for their contributions, forming a virtuous cycle.

- Contribution Value Quantification
- Incentive Distribution Management
- Points Exchange System
- Transparent Settlement Mechanism

Multi-win Value Network

Open Source Developers

Technical capabilities precisely quantified and recognized

- Multi-dimensional Capability Profiles & Rankings
- Transparent Contribution Value Display
- Receive Ecosystem Incentive Rewards
- Connect with Quality Career Opportunities

Enterprises & Organizations

Efficiently discover and connect with target talents

- Precise Talent Insights & Assessment
- Targeted Developer Outreach
- Incentivize Core Contributors
- Enhance Open Source Influence

Open Source Communities

Build healthy and sustainable ecosystems

- Community Contribution Data Analysis
- Core Member Identification
- Contributor Incentive Mechanism
- Ecosystem Health Monitoring

Let every open source contribution
receive its deserved value







CCOSI

